

MUSIC PERFORMANCE & STREAMING ANALYTICS

USING MS EXCEL AND POWER BI

1.PROJECT OVERVIEW

The music industry generates massive amounts of streaming data across digital platforms such as Spotify, YouTube, TikTok, Apple Music, Pandora, Deezer, Amazon Music, and SoundCloud. This project transforms raw music streaming data into meaningful business insights through data cleaning, exploratory data analysis (EDA), and interactive dashboard development using Microsoft Excel and Power BI.

2. OBJECTIVES

- Clean and standardize the raw music dataset.
- Handle missing values and duplicate records.
- Validate track, album, and artist metadata using ISRC where possible.
- Analyze artist performance, song performance, platform engagement, trends, and relationships.
- Build an interactive Power BI dashboard.
- Generate business insights and recommendations.

3. DATA SOURCE

Dataset: Music Performance & Streaming Analytics

Domain: Music Streaming Analytics

Dataset Size: Approximately 4,600 songs (final cleaned dataset: 4,598 songs), 27 columns.

Platforms: Spotify, YouTube, TikTok, Apple Music, Deezer, Pandora, Amazon Music, SoundCloud, and Shazam.

4. PROBLEM STATEMENT

The music industry releases thousands of songs across multiple streaming services. This project identifies top-performing artists, songs, albums, platform contributions, release trends, and relationships between streaming metrics to support data-driven decision-making

5. DATASET UNDERSTANDING

Column Name	Data Type
Track	Text
Album Name	Text
Artist	Text
Release Date	Date
ISRC	Text
All Time Rank	Whole Number
Track Score	Decimal Number
Spotify Streams, Spotify Playlist Count, Spotify Playlist Reach, Spotify Popularity	Whole Number
YouTube Views, YouTube Likes, YouTube Playlist Reach	Whole Number
TikTok Posts, TikTok Likes, TikTok Views	Whole Number
Deezer Playlist Count, Deezer Playlist Reach	Whole Number
Pandora Streams, Pandora Track Stations	Whole Number
Apple Music Playlist Count, Air Play Spins, SiriusXM Spins, Amazon Playlist, Soundcloud Streams, Shazam Counts	Whole Number

6. DATA CLEANING & PRE-PROCESSING

- **Removed duplicate records**

The screenshot shows a Microsoft Excel spreadsheet with a data table. The columns are labeled A through N, corresponding to the following headers: Track, Album, Artist, Release Date, NSRC, All Time, Track ID, Spotify, Spotify Playlist, Spotify Popularity, YouTube Views, YouTube Likes, TikTok Posts, and TikTok Likes. The data rows are numbered 1 through 4603. A dialog box is open in the center of the screen, displaying the message: "2 duplicate values found and removed; 4596 unique values remain." with an "OK" button.

• Handled missing values

Album Name	Artist	Release D	ISRC	All Time R	Track Scor	Spotify Sh	Spotify Pl	Spotify Pc	YouTube I	YouTube T	TikTok Po	TikTok Lik	TikTok Vie	YouTube f	Apple Mu	AirPlay Sp	SiriusXM	Deezer Pl	Deezer Pl	Amazon P	Pandora S	Pandora T	Soundclo	Shaz	
Million Dollar Baby - Single	Tommy Richman	#####	QM24S24K	1	725.4	39,04,70,9	30,716	19,66,31,5	92	8,42,74,75	17,13,126	57,67,700	65,15,65,9,5,33,22,81	15,05,97,0	210	40,975	684	62	1,75,98,71	114	1,80,04,65	22,931	48,18,457	26,69	
Not Like Us	Kendrick Lamar	5/4/2024	USUG1240	2	545.9	32,37,03,8	28,113	17,45,97,1	92	11,63,47,0	94,86,739	6,74,700	3,52,23,54	20,83,39,0	15,63,80,3	188	40,778	3	67	1,04,22,43	111	77,80,028	28,444	66,23,075	11,18
I like the way you kiss me	Artemas	#####	QZJ84240X	3	538.4	60,13,09,2	54,331	21,16,07,6	92	12,25,99,1	22,28,730	30,25,400	27,51,54,2,3,36,91	20,37,37,84,9	190	74,333	536	136	3,63,21,84	172	50,22,621	5,639	72,08,651	52,85	
Flowers - Single	Miley Cyrus	#####	USSM122K	4	444.9	2,03,12,80	2,69,802	13,65,69,0	85	1,09,61,00	1,06,29,79	71,89,811	1,07,87,57	14,60,37,2	3,35,11,88	394	14,74,799	2,182	264	2,46,84,24	210	19,02,60,2	2,03,384	1,18,	
Houdini	Eminem	#####	USUG1240	5	423.3	10,70,34,9	7,223	15,14,69,8	88	7,73,73,95	36,70,188	16,400		11,27,63,8	182	12,185	1	82	1,76,60,62	105	44,93,884	7,006	2,07,179	4,57,	
Lovin On Me	Jack Harlow	#####	USAT2231	6	410.1	67,06,65,4	1,05,892	17,54,21,0	83	13,11,48,0	13,92,593	42,02,367	21,49,43,4	2,93,86,86	2,86,72,22	138	5,22,042	4,654	86	1,71,67,25	152	13,85,29,3	50,982	94,38,601	45,17
Beautiful Things	Benson Boone	#####	USWB123K	7	407.2	90,01,58,7	73,118	20,15,85,7	86	30,87,23,1	41,41,20,780		2,95,84,94	53,49,15,3	4,60,15,79	280	3,83,478	429	168	4,81,97,85	154	6,54,47,47	57,372	99,90	
Gata Only	FloppyMenor	2/2/2024	QZL38240N	8	375.8	67,50,79,1	40,094	21,12,36,9	92	22,83,82,5	14,39,495	35,00,000	33,85,46,6	3,80,45,84	2,11,25,81	160	17,221	30	87	3,32,45,59	53	33,72,428	5,762	60,63	
Don Omer	MUSIC LAB JPN	6/9/2024	TCJPA246J	9	355.7	1,65,30,18	1	15																	
BANDABAND (feat. Lil Baby)	Central Cee	#####	USSM124K	10	330.6	9,06,76,57	10,400	18,41,99,4	86	3,27,35,24	9,88,682	3,25,800	12,15,74,5	97,46,56,2	17,47,06,8	191	3,823	117	78	1,08,00,09	92	10,05,626	842	36,75,709	6,66,
I Had Some Help	Post Malone	#####	USUM724K	11	322.1	25,00,80,4	16,219	16,87,66,1	91	3,22,29,06	5,50,191		1,84,500	21,00,000	32,71,30,7	157	62,312	71	60	1,20,09,07	114	2,35,55,47	21,172	15,94,605	9,63,
The Door	Teddy Swims	#####	USWB124K	12	320.2	11,11,55,1		2,63,767																	
HIT ME HARD AND SOFT	Billie Eilish	#####	USUM724K	13	316.3	22,16,36,1	13,800	19,72,80,6	92	4,00,22,52	13,07,290	1,95,300	4,58,89,00	36,00,17,0	34,21,07,1	244	41,344	45	138	3,82,43,63	163	13,54,692	1,219	13,13,357	4,50,
WE DON'T TRUST YOU	Future	#####	USSM124K	14	308.2	35,51,86,4	43,025	10,81,68,6	88	9,80,81,49	14,56,177			17,44,91,4	153	90,237	3	55	30,01,826	109	2,62,52,26	18,364	1,20,38,03	19,63	
bathroom floor	Kids With Buns	#####	NLC24210	15	301.6	2,97,185	277	24,445	12	6,73,65,02	1,67,85,48	3	1,088	22,234		1	1,057								

All Time Rank	Track Score	Spotify Stream	Spotify Playlist Count	Spotify Playlist Reach	Spotify Popular	YouTube Views	YouTube Likes	TikTok Post	TikTok Likes	TikTok Views	YouTube Playlist Reach	Apple Music Playlist Count	AirPlay Sp	SiriusXM Sp	Deezer Playlist Count
1	725.4	390470936	30716	196631588	92	84274754	1713126	5767700	651565800	5332281936	150597040	210	40975	684	62
2	545.9	313703884	28113	174597137	92	116347040	9486739	674700	35212547	208339015	156380351	188	40778	3	67
3	538.4	601309183	54331	211607669	92	122599116	2228730	3025400	275154237	9369120610	973784955	190	74333	536	136
4	444.9	2031280633	269802	136569078	85	1096100899	10629796	7189811	1078757968	14603725994	5351188582	394	1474799	2182	264
5	423.3	107034922	7223	151469874	88	77373957	3670188	16400	26576922	266064027	112763851	182	12185	1	82
6	410.1	670665438	105892	175421034	83	131148091	1392593	4202367	214943489	2938686633	2867222632	138	522042	4654	86
7	407.2	900158751	73118	201585714	86	308723145	4120760	182529	29584940	534915313	4601579812	280	383478	429	168
8	375.8	675079153	40094	211136940	92	228381568	1439495	3500000	338546668	3804584163	2112581620	160	17221	30	87
9	355.7	1653018119	1	15	63	148269610	1257935	182529	26576922	266064027	97746040	0	5954	86	0
10	330.6	90676573	10400	184199419	86	31735244	988682	315800	121574500	974656200	174706874	191	3823	117	78
11	322.1	250080413	16219	168766158	91	32229063	550191	182529	184500	2100000	327130768	157	62312	71	60
12	320.2	1111155129	2	263767	63	148269610	1257935	182529	26576922	266064027	97746040	0	5954	86	0
13	316.3	221636195	13800	197280692	92	40022524	1307290	195300	45889000	360017000	342107131	244	41344	45	138
14	308.2	355186436	43025	108168634	88	98081493	1456177	182529	26576922	266064027	174491415	153	90237	3	55
15	301.6	297185	277	24445	12	6736502312	16785489	3	1088	22234	97746040	1	1057	86	0
16	299.9	915655569	103605	79944921	79	994302031	4294166	11054500	521725116	7499234052	2042145659	94	92331	228	60
17	297.6	395433400	12784	177932568	89	104673636	2484063	139300	25348800	207317000	268893710	221	129968	3	99
18	293.7	1258569934	105218	198645580	87	141646818	1514837	2294429	419319161	5456156211	2667285437	207	667329	3310	188
19	292.6	91272461	6499	52287548	86	16038053	569377	941900	119740900	1225345800	5839168	76	181	1	24
20	292.0	3301814535	407119	169783402	87	800899315	9101589	2755903	1633970617	15405881843	7289707052	455	1777811	4909	330
21	290.5	1065399331	170890	121712197	78	240970670	2749668	3091732	652428798	7095564106	1063591802	158	400899	273	143

• Corrected metadata using ISRC where available

Artist	ISRC
	QZNRWQ2410638
	QZRYFZ2445017
	QZNRWT2471497
	QZNRWU2402635
	MX2832415361

7. TOOLS & TECHNOLOGIES

Excel: Data Cleaning and Transformation.

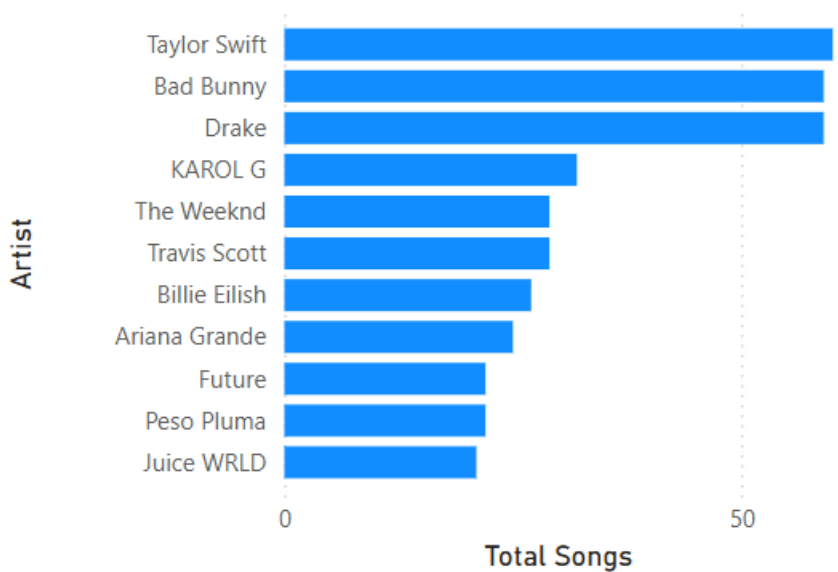
Power BI: Data Modelling, DAX Calculations, Visualization, And Interactive Dashboard Creation.

8. EXPLORATORY DATA ANALYSIS (EDA)

EDA included:

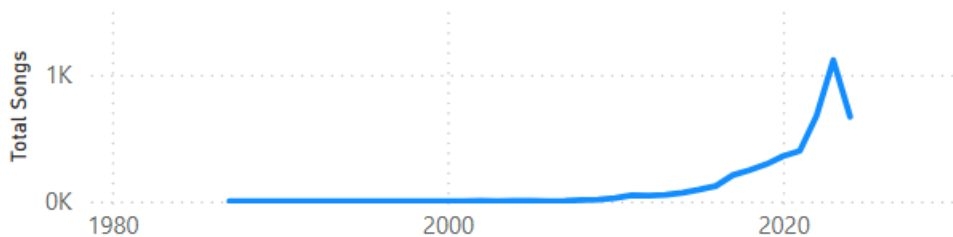
- Distribution of artists and song

Total Songs by Artist



- Release year analysis

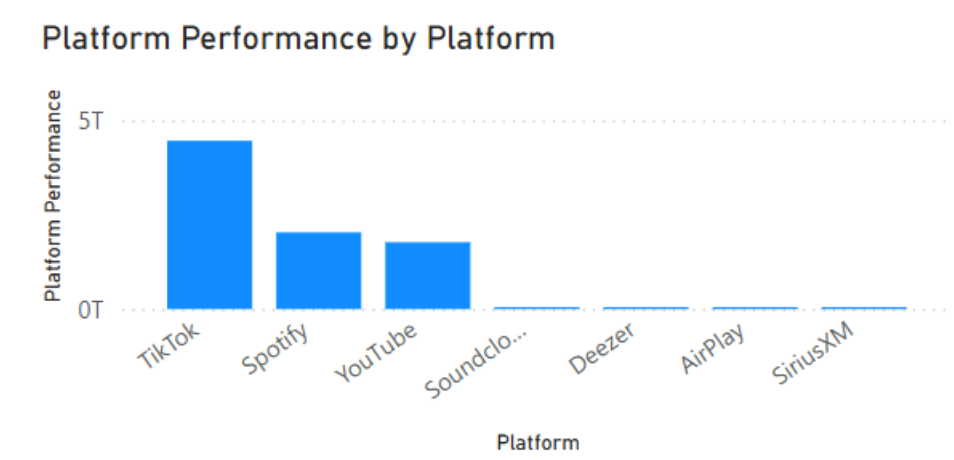
Total Songs by Release Year



2023 has the highest number of songs released in the dataset.

Total Songs - 1115 songs.

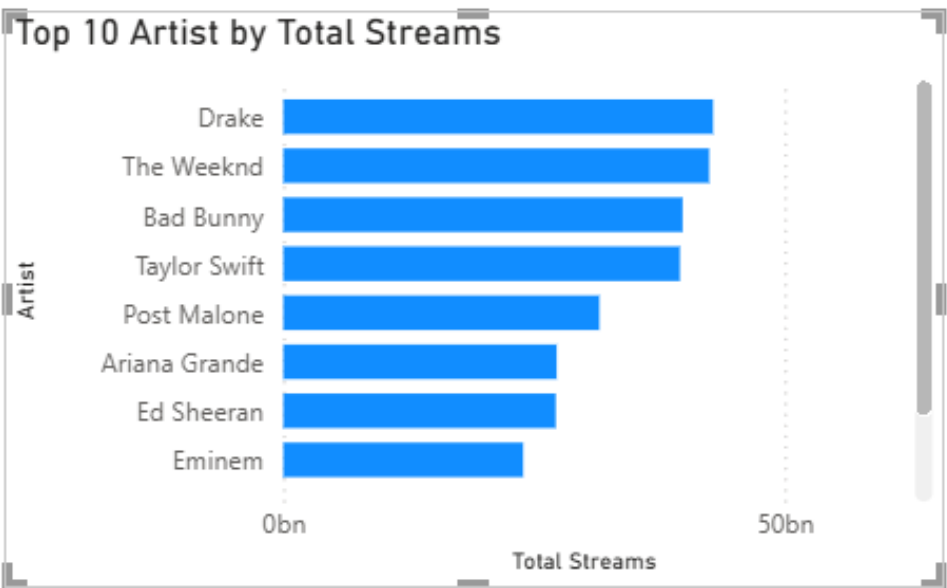
• Platform statistics



TikTok Platform contributes the most to song popularity.

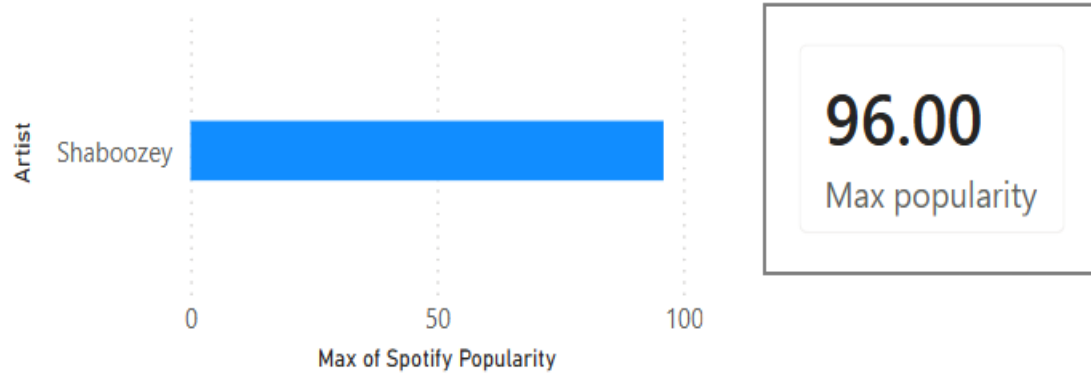
9. ARTIST PERFORMANCE ANALYSIS

• Top 10 artists by total streams



- Highest popularity artists

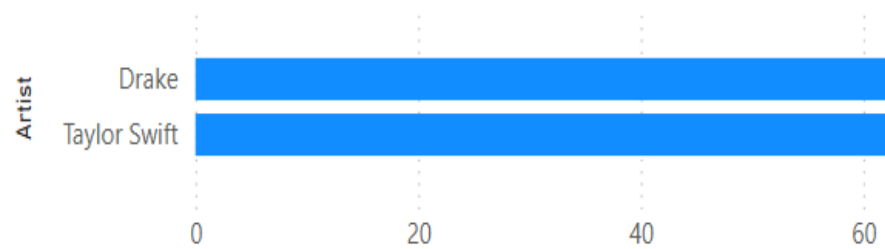
Highest Popularity by Artist



Shaboozey
Highest Popularity of 96

- Most frequently appearing artists

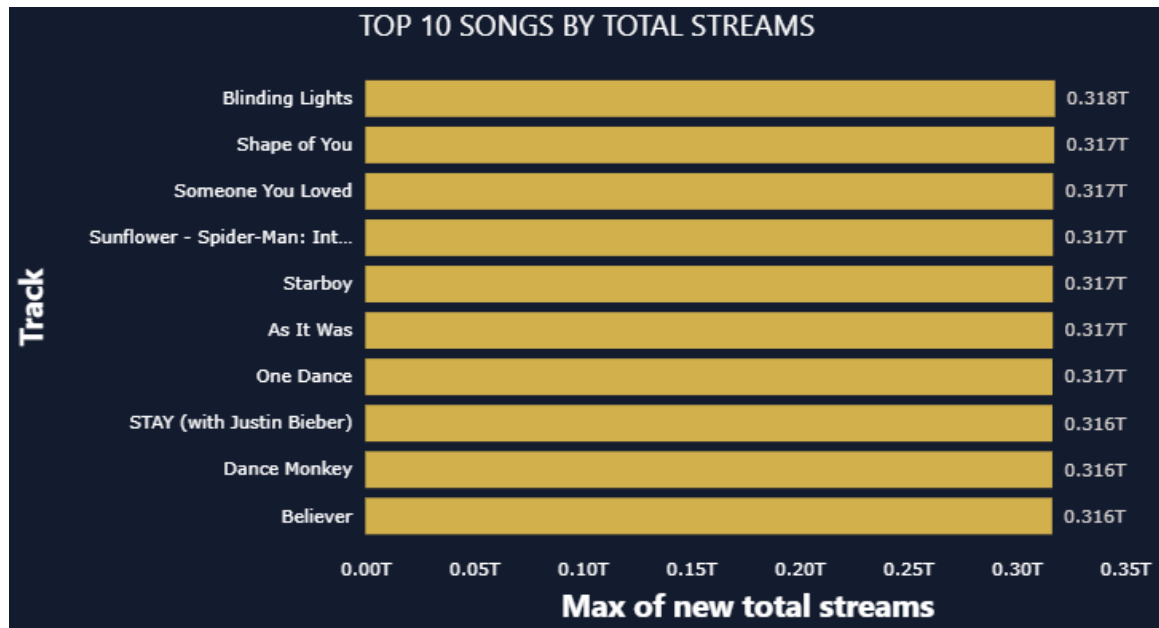
Most Frequently Appearing Artist



Drake and Taylor Swift appear most frequently in the dataset, with 63 songs in each.

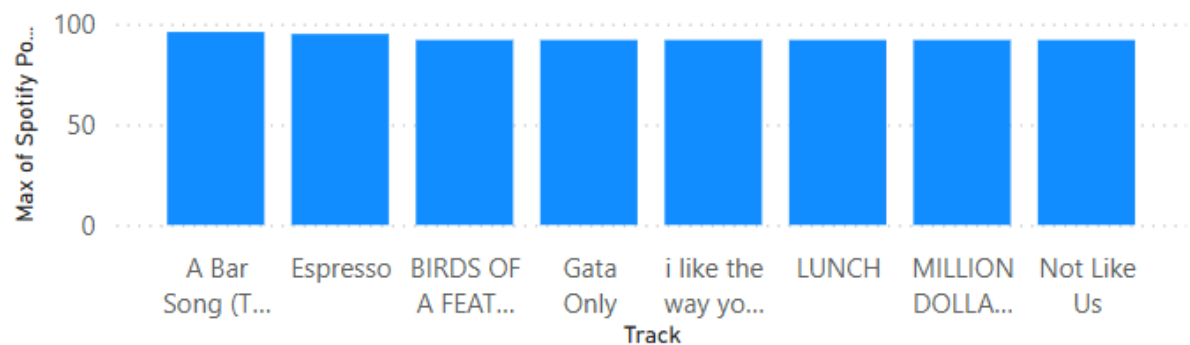
10. SONG PERFORMANCE ANALYSIS

- Top streamed songs

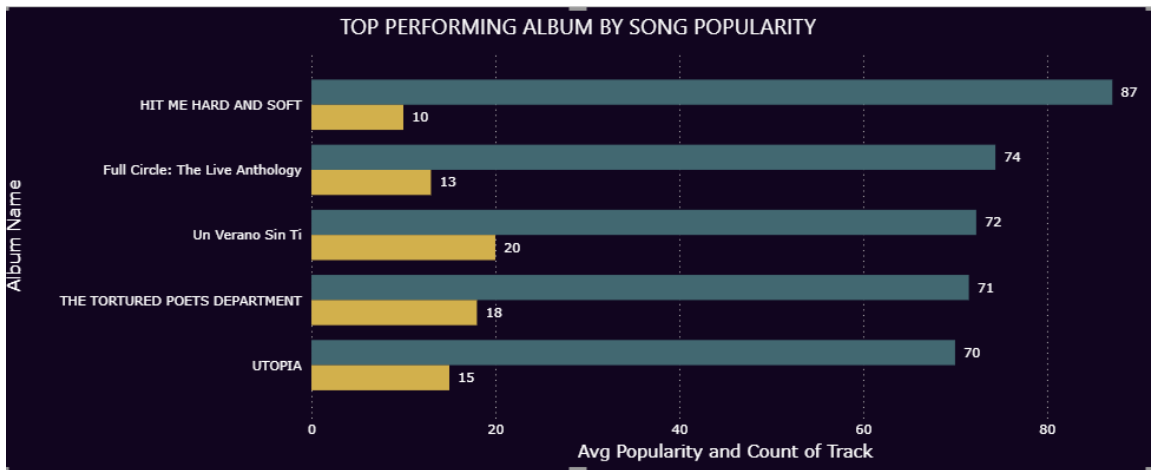


- Most popular songs

Highest Popularity of Songs



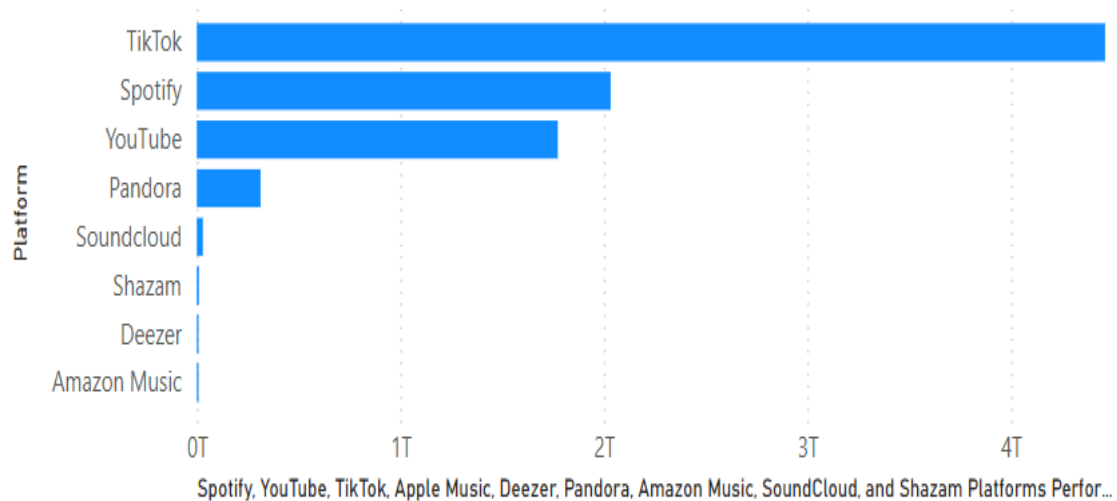
- Albums containing the highest-performing tracks



11. PLATFORM ANALYSIS

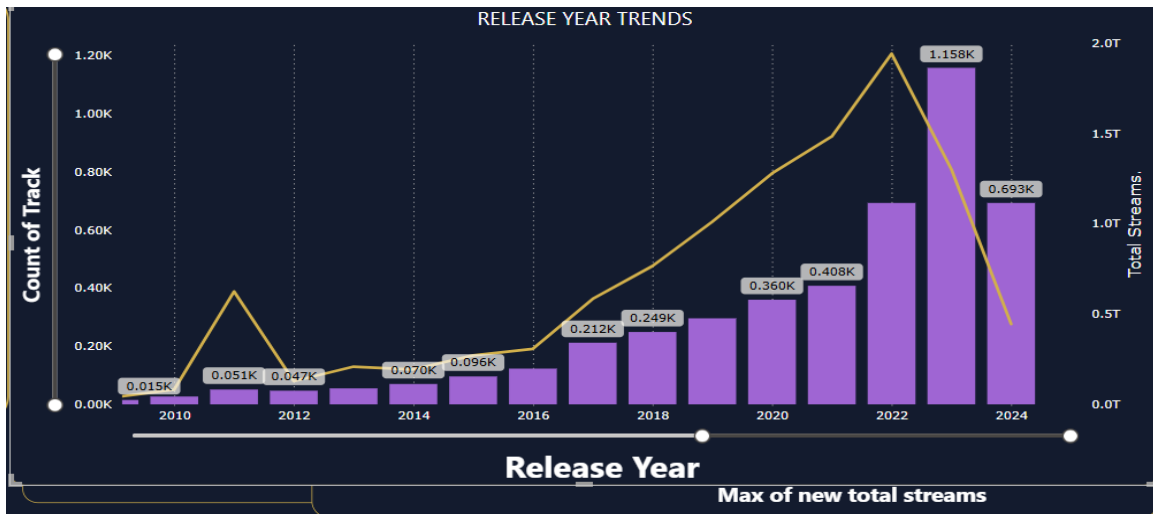
Compared Spotify, YouTube, TikTok, Apple Music, Deezer, Pandora, Amazon Music, SoundCloud, and Shazam.

Spotify, YouTube, TikTok, Apple Music, Deezer, Pandora, Amazon Music, SoundCloud, and Shazam Platforms Performance by Platform



12. TREND ANALYSIS

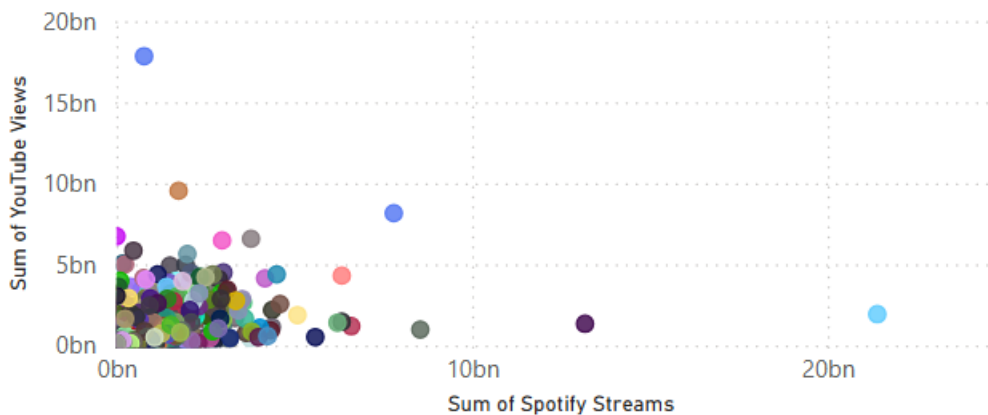
Analyzed release-year trends



13. RELATIONSHIP ANALYSIS

• Spotify Streams vs YouTube Views

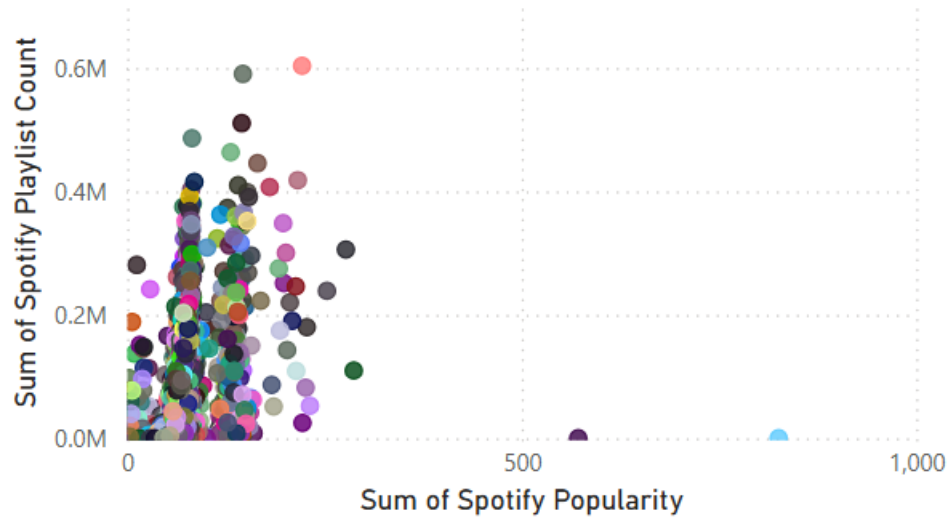
Sum of Spotify Streams and Sum of YouTube Views by Track



There is a moderate positive relationship between Spotify streams and YouTube views. Most songs with higher Spotify streams also have higher YouTube views.

- **Spotify Popularity vs Playlist Count**

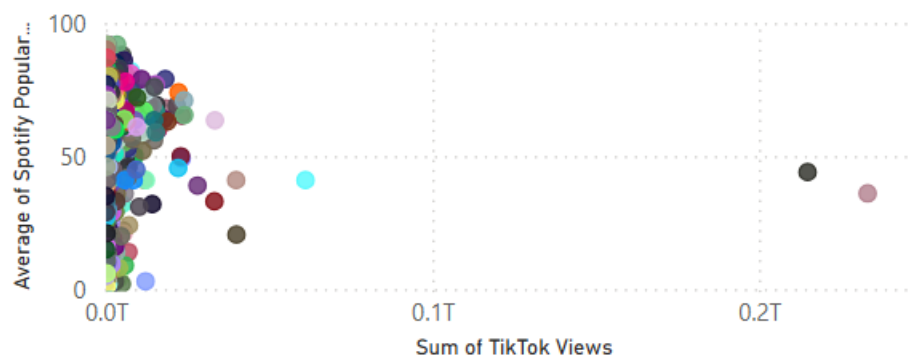
Sum of Spotify Popularity and Sum of Spotify Playlist Count by Track



Songs with higher playlist counts generally have higher Spotify popularity. This indicates that playlist placements can contribute to increasing a song's popularity.

- **TikTok Views vs Spotify Streams**

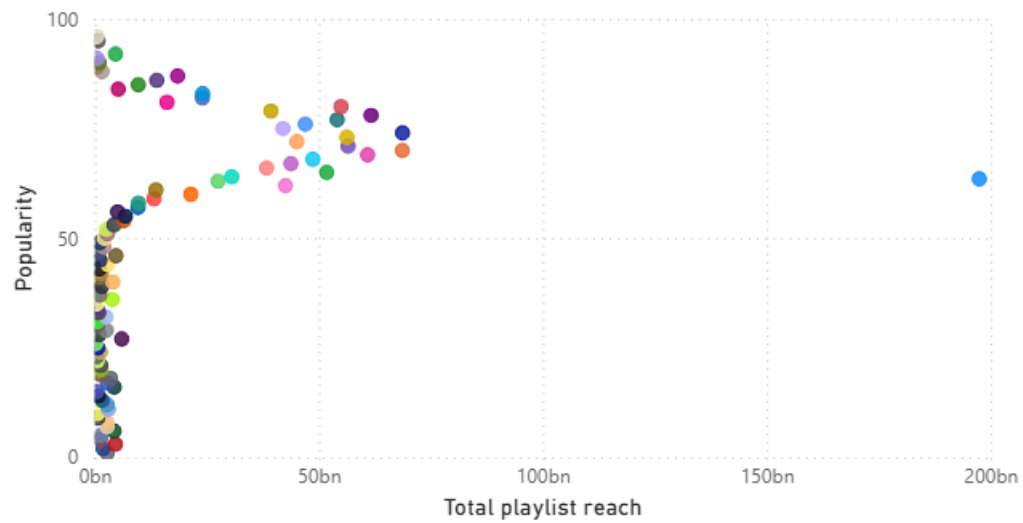
Sum of TikTok Views and Average of Spotify Popularity by Track



TikTok views have a moderate impact on Spotify popularity. Some songs with more TikTok views have higher Spotify popularity, while other.

- **Playlist Reach vs Popularity**

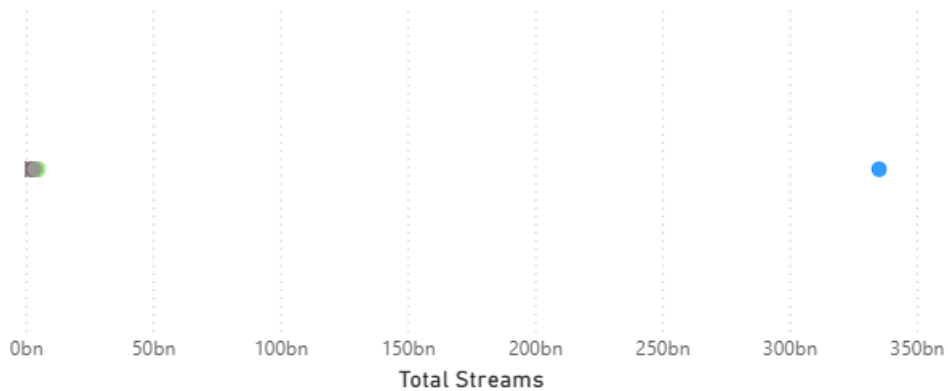
Total playlist reach by Popularity



Higher playlist reach generally leads to higher song popularity.

- **Shazam Counts vs Streaming Performance**

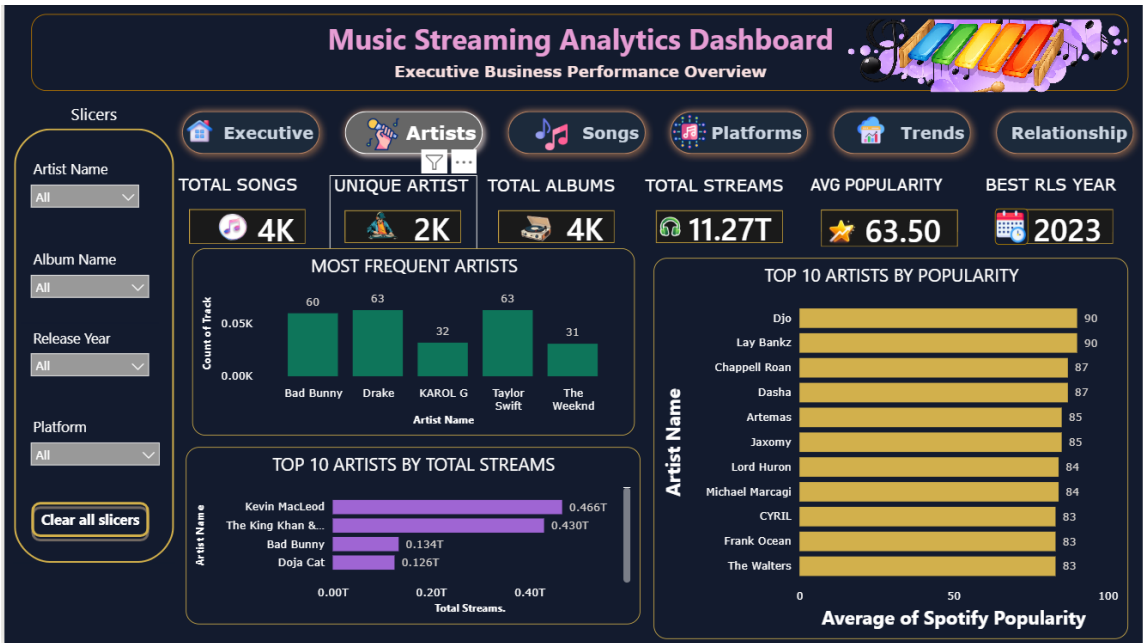
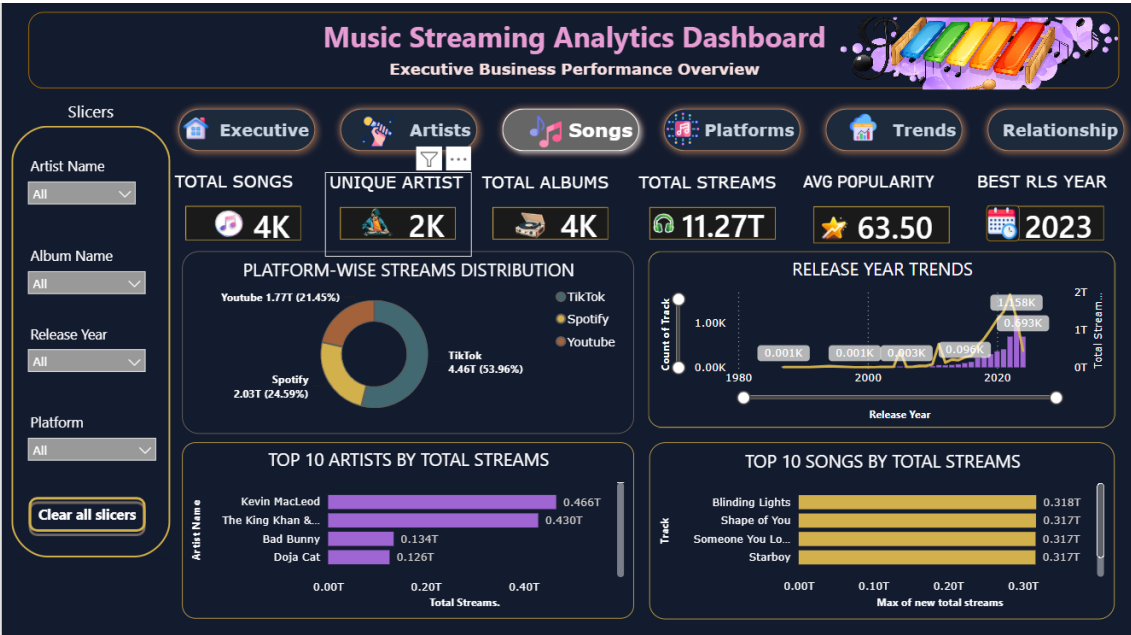
Total Streams by Shazam Counts

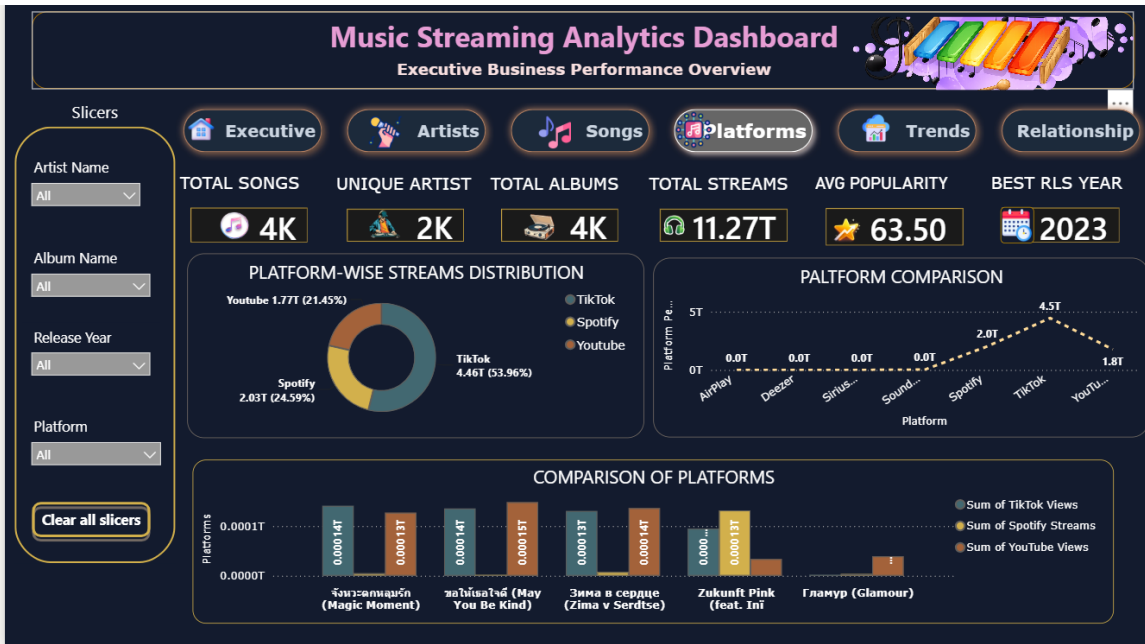
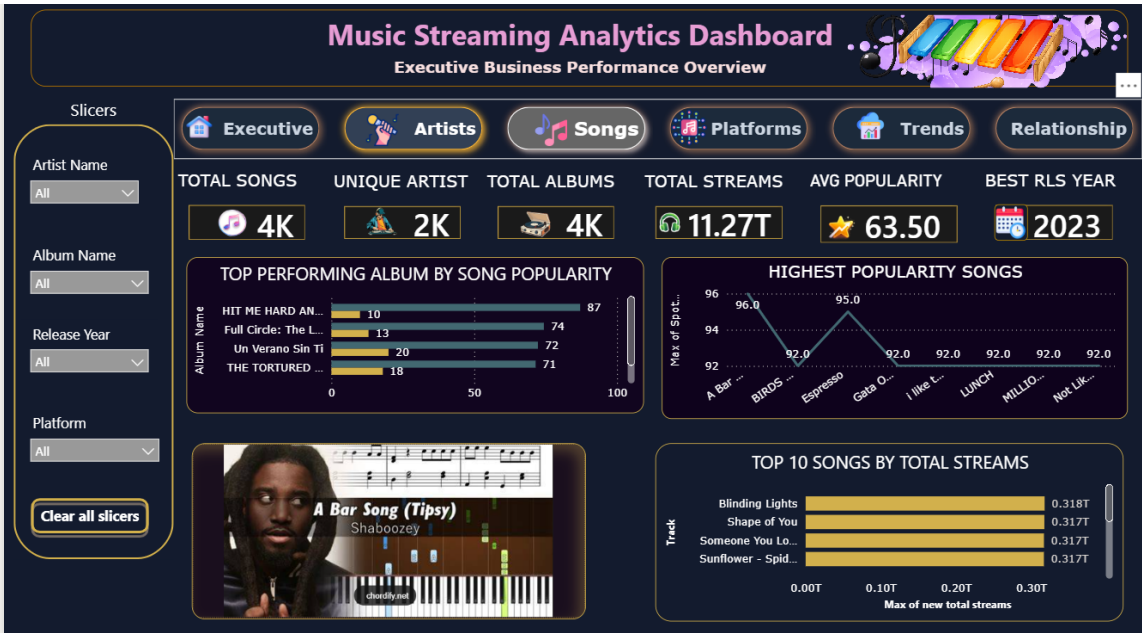


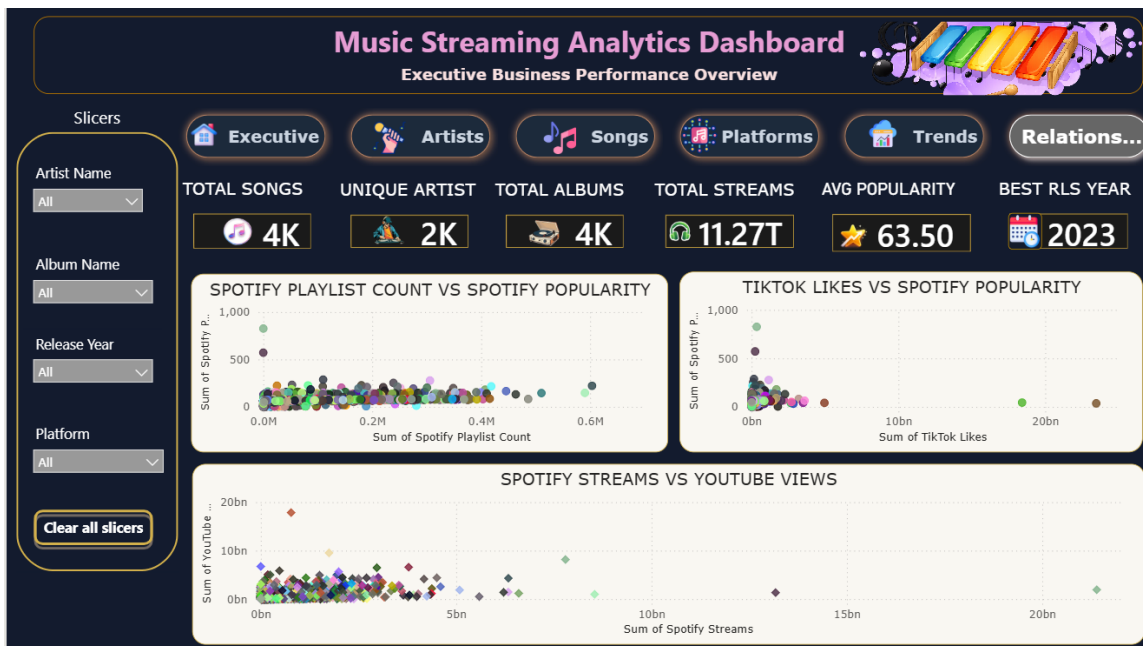
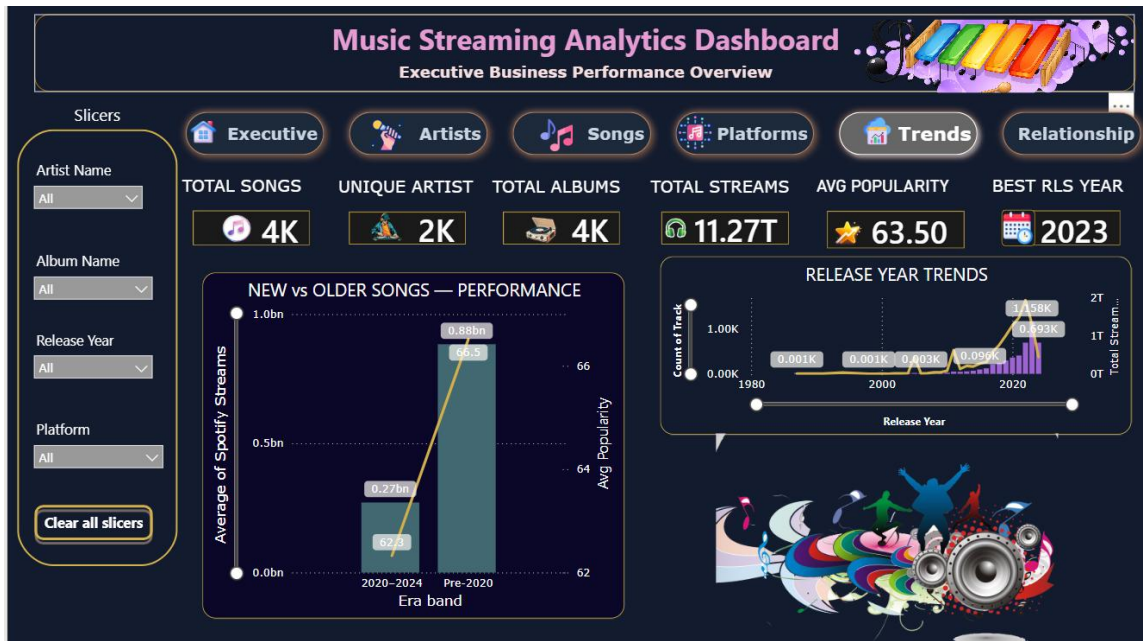
Higher streaming performance is associated with higher shazam counts.

Strong positive relationships were observed between playlist exposure and streaming success.

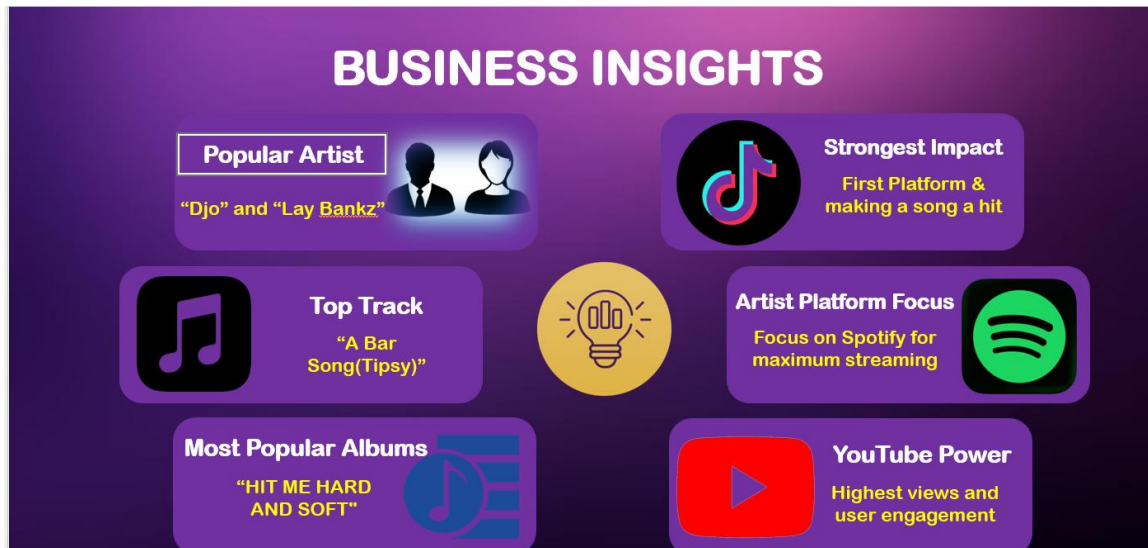
DASHBOARD PREVIEW







15. Business Insights



16. Recommendations



18. CONCLUSION

This project successfully transformed raw music streaming data into actionable business insights using Excel and Power BI. The dashboard supports data-driven decisions for artists, record labels, and streaming platforms.